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BITS Pilani
Pilani Campus

CE F230

Civil Engineering Materials

Chemical admixtures

Admixtures



A material other than water, aggregate, cement and reinforcing fibers that is used in concrete as an ingredient, and added to the batch immediately before or during mixing

Admixtures



- Why are they used?
 - To modify properties of fresh and hardened concrete
 - To ensure the quality of concrete during mixing, transporting, placing and curing
 - To overcome certain unexpected emergencies during concrete operations



Admixtures

❖ Classification

- ✓ Chemical admixture
- ✓ Mineral admixture

❖ Admixtures exclude essential concrete ingredients

❖ They should be used cautiously and for good reason

Chemical Admixtures for Concrete



- Added in small amounts to the concrete (usually no larger than 5% by weight of cement) and dissolved in mixing water
- Types of admixtures
 - Air-entraining admixtures (ASTM C260)
 - Water-reducing admixtures (ASTM C494 and 1017)
 - Water reducers
 - Superplasticizers
 - Set-controlling admixtures (ASTM C 494)
 - Accelerators
 - Retarders
 - Miscellaneous admixtures



Air-entraining Admixtures

- These admixtures produce tiny, dispersed air bubbles (0.05 to 1.25 mm) into the concrete
 - Water expands 9% as it freezes causing internal stress that cracks the hardened cement paste and greatly reduces durability
 - Air entrainer provides space for the water to go as it expands
- Recommended for **all concrete exposed to freezing**



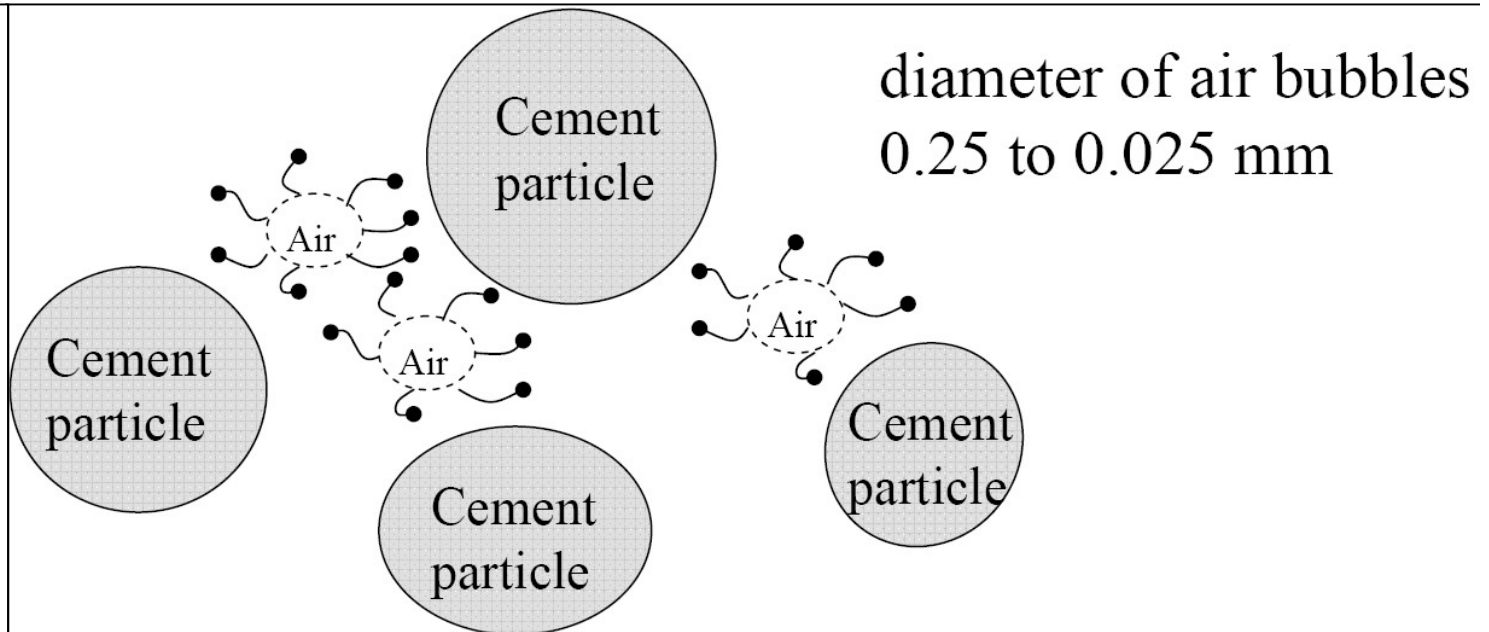
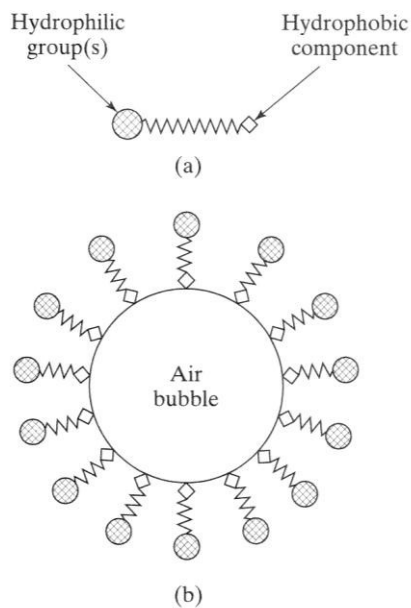
Air-entraining Admixtures

- **Improve workability**, resistance to freeze-thaw cycles, de-icing chemicals, sulfates, & alkali-silica reaction
- Decreases strength (1% air causes 5% loss in strength) but can be compensated with **lower w/c ratio**

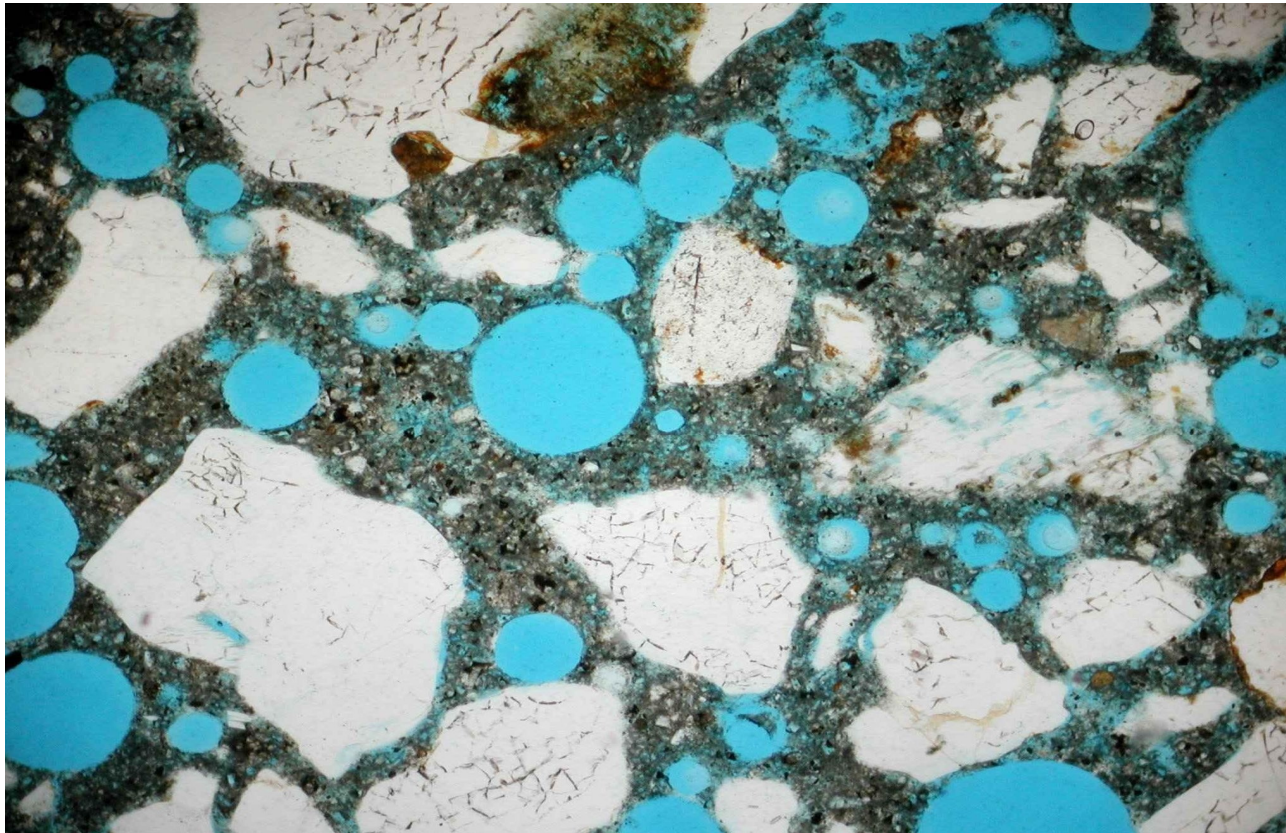


Mechanism of Air Entraining

- Air-entraining admixtures are surface-active agents which act at the air-water interface, causing water to foam during mixing, similar to *detergents*
- The fine foam remains stable until it is “locked” into the cement paste during hardening



Thin section of concrete with air voids dyed blue



Different from entrapped air which is harmful larger bubbles

- Frost resistance improves with decreasing void size
- Small voids reduce strength less than large ones

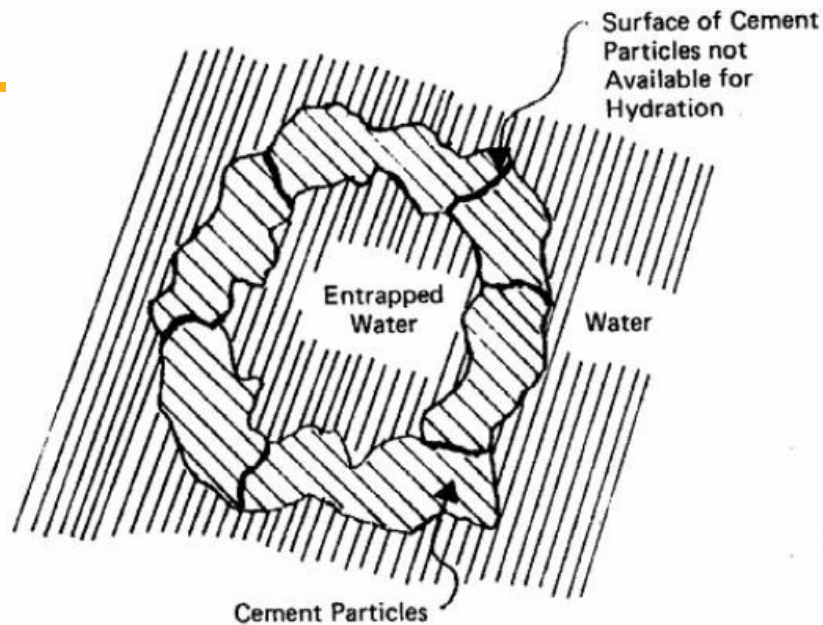


Water Reducers (Normal and Mid-Range)

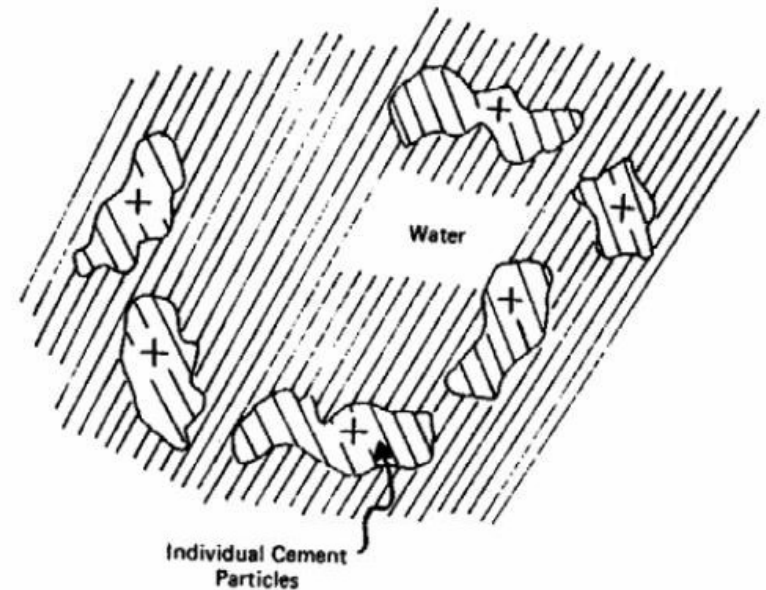
- These admixtures lower the water required (8-12% reduction) to attain a given slump
- Used to
 - Improve workability at same w/c ratio
 - Increase strength at same workability
 - Reduce cost at same w/c ratio and workability

(Source: Hewlett 1978)	Cement content (kg/m ³)	Water/Cement Ratio	Slump (mm)	Compressive Strength (MPa)	
				7 day	28 day
Base mix	300	0.62	50	25	37
Improve workability	300	0.62	100	26	38
Increase strength	300	0.56	50	34	46
Reduce costs	270	0.62	50	25.5	37.5

Mechanism of Water Reduction



Cement particles in the absence of dispersing agent



Effect of dispersant on the cement-water mixture

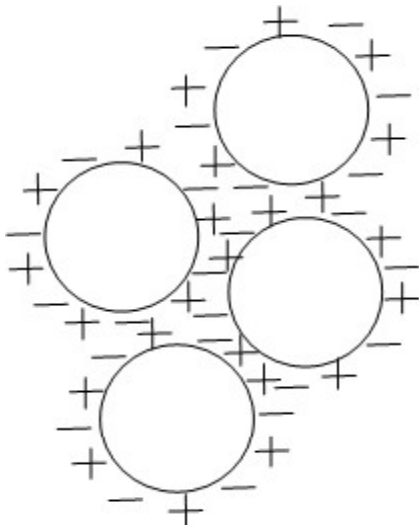
Addition of water reducer results in:

- Liberating water entrapped by surrounding cement particles so that it can contribute to fluidity of concrete
- Making additional surfaces of cement particles available for early hydration

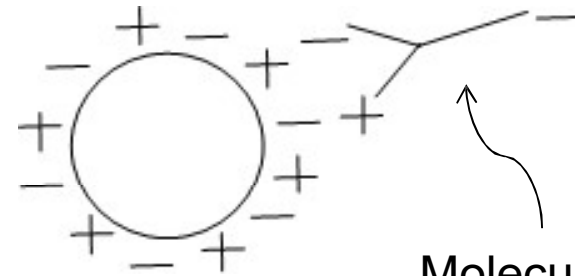
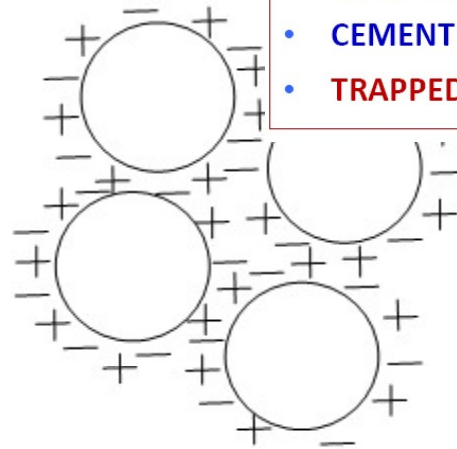


Mechanism of Water Reduction

- CEMENT PARTICLES WITH EITHER +VE OR -VE ELECTROSTATIC CHARGE
- WATER TRAPPED WITHIN CEMENT PARTICLE GROUP
- WATER-REDUCER PROVIDE EQUAL POLARITY TO CEMENT PARTICLES
- CEMENT PARTICLES GET DISPERSED
- TRAPPED WATER RELEASED THUS INCREASING WORKABILITY

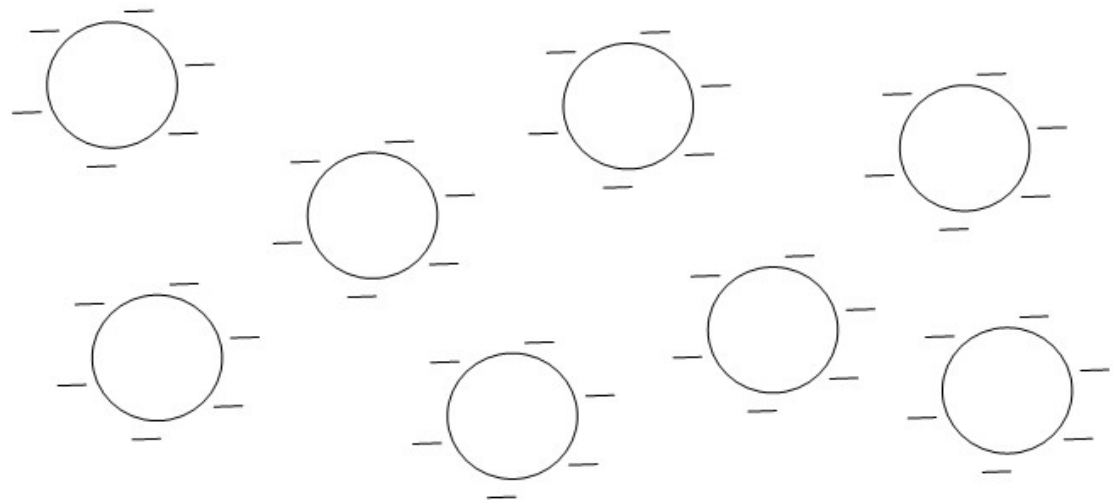


Without Water Reducer: Unlike charges attract, causing cluster of cement grains to flocculate



Molecule of Water Reducer

With Water Reducer: Repulsion of like charges pushes the cement grains apart



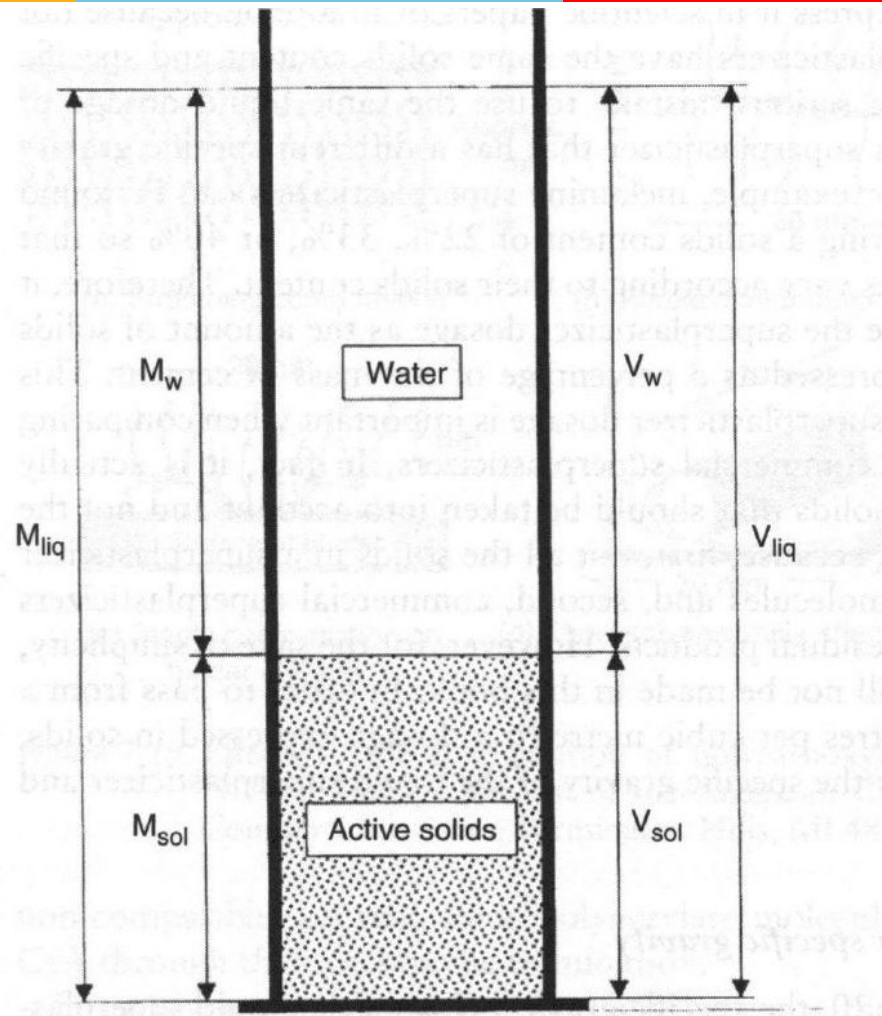
Superplasticizers (High-Range Water Reducers)



- These are a more recent and more effective type of water reducer (12-40% reduction)
- Used to produce **flowing concrete** in situations where placing in inaccessible locations or where very rapid placing is required
- Also used to produce **very-high strength concrete**, using normal workability but a very low w/c



Superplasticizer



SCHEMATIC REPRESENTATION OF A SUPERPLASTICIZER



Practical Use of Superplasticizers

- For increasing the workability of mixes, normal dosage of superplasticizer is **1-3 liters/m³** of concrete
- Liquid superplasticizer contains **~40% of active material**
- Concentration of solids in commercial superplasticizers vary
- Any comparison of performance must be made on the basis of the **amount of solids** and not on the total mass
- Volume of liquid superplasticizer must be taken into account when calculating the w/c and mix proportions in general

Expressing the dosage:

- Liters per m³ of concrete
- Solid content as a percentage of the mass of cement:
 - **Solid content, s (%) = $[M_{\text{sol}} / M_{\text{liq}}] / 100$**
 - **Mass of solid = $s \times G_{\text{sp}} \times V_{\text{liq}}$**



Accelerators

- These admixtures accelerate the setting, hardening or the development of early strength of concrete
- Used to
 - Reduce the amount of time before finishing operations begin
 - Reduce curing time
 - Increase rate of strength gain
- Applications
 - Urgent repair work
 - Early formwork removal (productivity)
 - **Cold-weather concreting**

Calcium chloride (CaCl_2) is most common accelerator used



Retarders

- These admixtures delay or retard initial set
 - Slow down the early hydration reaction
 - Usually do not reduce final setting time much
 - May reduce early strength
- Applications
 - **Hot weather concreting:** the normal setting time is shortened by higher temperature
 - Unusual placement
 - Long haul distance
 - Special finishes (e.g., exposed aggregate)
- Sugar is a cheap retarder



Thank you